

6buwan: A RAW-vs-HOG Pipeline for One-Class SVM Signature Verification

ogcz

Abstract. A bank keeps genuine specimen signatures for a depositor but never a forgery in advance, which rules out ordinary two-class classification and forces a one-class formulation: a model fit only to a writer's genuine samples that still has to flag forgeries it has never seen. Histogram of Oriented Gradients (HOG) was designed to tolerate the kind of variation — stroke thickness, small shifts, local contrast — that raw pixel intensity is sensitive to, so intuition favors it as a feature representation. **6buwan** is a small, reproducible pipeline built to test that intuition rather than assume it: a One-Class SVM (OCSVM) is trained per writer on raw-pixel features and, separately, on HOG features, under an identical preprocessing and evaluation protocol, and the two are compared on CEDAR (locked, primary result) and on UTSig (a cross-dataset robustness check). Across six metrics and two threshold-independent checks (ROC-AUC, Matthews Correlation Coefficient), there is no universal winner: HOG's edge on CEDAR is not statistically significant at $n = 55$ writers and shrinks as training data grows, while on UTSig the two representations trade the lead depending on which metric and operating point is used. The result is a cautionary one for practitioners: whether a hand-designed feature "helps" a one-class verifier is dataset- and metric-dependent, not a settled question.

1 Introduction

Offline handwritten signature verification is, in deployment, a one-class problem. An institution such as a bank enrolls a depositor's genuine signature specimens; it does not, and cannot, enroll examples of how that depositor's signature might later be forged. A verifier that needs both classes at training time is answering a question the deployment setting never actually poses. The One-Class Support Vector Machine (OCSVM) [2] is built for exactly this setting: it learns a boundary around a single class and flags anything — including a skilled forgery it has never seen — that falls outside it.

Given that constraint, a natural question follows: is it worth extracting a hand-designed feature such as Histogram of Oriented Gradients (HOG) [3] at all, or does a one-class model do just as well working directly on raw pixel intensity? HOG was designed to tolerate stroke-thickness, small-shift, and local-contrast variation that raw pixels are sensitive to, so intuition favors it. Prior work on curvelet-based OCSVM signature verification [4] shows engineered features can help in a related setting, but under a different protocol than the one used here. **6buwan** was built to test the raw-vs-HOG question directly, under a genuinely one-class, low-training-data regime, rather than assume HOG wins going in — the guiding question throughout is whether HOG *affects* OCSVM performance relative to a raw-pixel baseline, not whether it improves it.

2 Related Work

Support Vector Machines [1] and their one-class extension [2] provide the verification model used throughout. HOG [3], originally developed for pedestrian detection, is used here purely as a feature representation. Kalera et al. introduced the CEDAR signature corpus used as the primary dataset [5]. Guerbai et al. evaluated OCSVM with curvelet features for signature verification [4] and is the closest prior comparison point, though under a different, non-identical protocol, so this paper treats it as context rather than a controlled baseline. General classifier-evaluation practice (balanced accuracy under class imbalance, ROC-AUC as a threshold-independent check) follows standard treatment [6].

3 Pipeline

Figure 1 shows the four pipeline stages, run identically for both feature representations so that any performance gap traces back to the representation itself and not the surrounding plumbing: preprocess every scan the same way, split each writer's signatures into train / validation / final-test, extract RAW or HOG features from the same preprocessed image, then fit and score one OCSVM per writer.

3.1 Preprocessing

An Otsu threshold finds the ink. A density-thresholded crop — a row or column only counts if it holds at least 5% of the busiest row/column's ink — removes blank margin without letting a thin trailing flourish drag the crop into a sliver. A hard aspect-ratio cap ($\leq 2.5{:}1$) pads anything still too elongated before the final resize to 100×100. Output stays grayscale rather than being re-binarized, so HOG still sees soft gradient information at stroke edges.

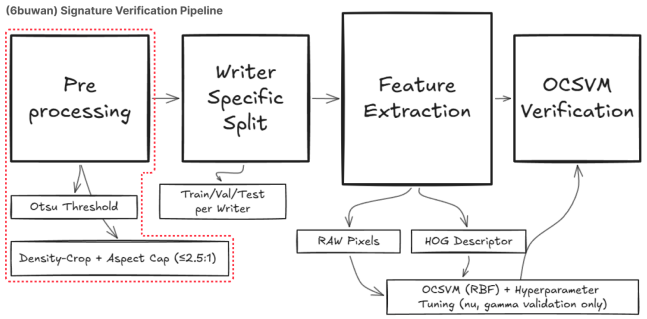


Figure 1: The 6buwan pipeline. Preprocessing (dashed) is shared by both feature branches.



Figure 2: Original scans (left of each pair) against their preprocessed, Otsu-cropped 100×100 output (right).

```
if width / height > max_aspect:
    # ceil, not floor -- floor can leave
    # the ratio slightly over the cap
    target_h = ceil(width / max_aspect)
    pad = max(0, target_h - height)
    img = np.pad(img, ((pad // 2,
        pad - pad // 2), (0, 0)),
        constant_values=0)
```

3.2 Feature Representations

RAW keeps absolute pixel intensity: a signature's exact position and shading is the whole signal. HOG discards that in favor of local gradient direction, on the premise that stroke direction survives shifts and thickness changes that sink raw pixels.

```
def create_raw_features(paths):
    return np.array([
        preprocess_signature(p)
          .astype(np.float32)
          .flatten() / 255.0
        for p in paths])

def create_hog_features(paths):
    return np.array([
        hog(preprocess_signature(p)
              .astype(np.float32) / 255.0,
            orientations=9,
            pixels_per_cell=(8, 8),
            cells_per_block=(2, 2),
```

```
            block_norm="L2-Hys")
        for p in paths])
```

3.3 Verification Model

One RBF-kernel OCSVM is fit per writer, on that writer's genuine signatures only:

```
model = OneClassSVM(
    kernel="rbf", nu=nu, gamma=gamma)
model.fit(X_writer_train)
```

`nu` and `gamma` are each swept over a fixed multiplier grid ($0.01\times$–$100\times$) centered on a data-driven reference gamma, $1/(n \cdot v)$ for feature count n and variance v; RAW and HOG each get their own reference, computed from their own dimensionality. Hyperparameters are tuned once per representation on the validation set only, then locked before the final-test data is touched at all.

4 Experimental Setup

4.1 Datasets

CEDAR — 55 writers, 24 genuine + 24 forged signatures each. Per writer: 12 genuine for training, 3 genuine + 12 forged for validation, 3 genuine + 12 forged held out for final testing. Forgeries are never used for training, consistent with the one-class formulation.

UTSig — 115 writers, 27 genuine + 6 skilled forgeries each, used as a cross-dataset robustness check run only after CEDAR's pipeline and hyperparameters were locked. Chosen because it is the only major CEDAR alternative not capped at 24 genuine signatures per writer. Genuine signatures split 21/3/3; skilled forgeries split 3/3 across validation/final-test, with none held out for training.

4.2 Evaluation

False Accept Rate (FAR), False Reject Rate (FRR), Accuracy, Precision, Recall, and F1 are computed per writer and aggregated, and checked against reject-all / accept-all trivial baselines and Balanced Accuracy, since CEDAR's final-test set skews roughly 80% forged. The UTSig run additionally reports Matthews Correlation Coefficient and per-writer ROC-AUC computed from `decision_function` scores, to separate "is this representation worse" from "is it worse only at the specific operating point the locked `nu` happens to pick." A paired Wilcoxon signed-rank test is used on both datasets to check whether the aggregate RAW-vs-HOG gap is distinguishable from writer-to-writer sampling variability.

5 Results

Table 1 reports CEDAR's locked final-test comparison, averaged across all 55 writers.

HOG shows the better value on all six metrics, but the margin is modest and, per writer, only narrowly favors HOG: across the 55 writers, HOG wins on F1 for 16 writers, RAW for 12, and 27 tie (a high tie rate driven by each writer's final-test set holding only 3 genuine signatures, which quantizes FRR to one of four values). A paired Wilcoxon signed-rank test finds neither the F1 gap ($p = 0.38$) nor the FRR gap ($p = 0.49$) statistically significant at $n = 55$. A follow-up run giving each writer 18 training signatures instead of 12 further narrows the gap, with F1 converging to ≈ 0.41 for both representations.

Table 1: CEDAR final-test results (locked hyperparameters).

Feat.	FAR	FRR	Acc.	Prec.	Rec.	F1
RAW	0.009	0.848	0.823	0.330	0.152	0.198
HOG	0.000	0.836	0.833	0.436	0.164	0.235

Table 2: Winner by metric, CEDAR vs. UTSig.

	CEDAR (locked)	UTSig
By F1	HOG (0.235 vs 0.198)	RAW (0.565 vs 0.482)
By ROC-AUC	HOG (0.917 vs 0.866)	HOG (0.650 vs 0.583)
Significant?	no ($p > 0.05$)	yes ($p < 0.05$)

Table 2 summarizes the cross-dataset picture, including UTSig, where the ranking is not the same as on CEDAR.

On UTSig, RAW wins the locked operating point on F1, but HOG ranks signatures better overall by ROC-AUC, and there the RAW-vs-HOG gap *is* statistically significant on both F1 and FRR. So the two datasets do not even agree on which representation to prefer, let alone by how much.

6 Discussion

There is no universal winner. On CEDAR, HOG's advantage is consistent in direction across every metric reported but is small in absolute terms, not significant at the tested sample size, and shrinks further as training data grows from 12 to 18 signatures per writer — consistent with HOG's built-in invariances mattering most exactly when training data is scarcest. On UTSig, the picture inverts on the threshold-based metric (RAW wins F1) while HOG still ranks signatures better overall (ROC-AUC), and here the gap is significant. Taken together, whether a hand-designed feature such as HOG helps a one-class signature verifier is not a fixed property of the feature; it depends on the dataset, the amount of per-writer training data, and which metric and operating point the comparison is made at. A comparably important, dataset-independent finding is that both representations, on both datasets, sit well above trivial reject-all/accept-all baselines on Balanced Accuracy, confirming both are doing genuine verification work rather than degenerate class-majority prediction.

7 Conclusion and Future Work

Under a genuinely one-class, writer-dependent protocol, replacing raw-pixel features with HOG affects OCSVM signature-verification performance, but the direction and significance of

that effect is dataset-dependent rather than a settled win for HOG. Planned follow-ups include a direct, same-protocol comparison against curvelet-based features [4], a feature-fusion pipeline combining RAW and HOG rather than treating them as exclusive alternatives, replicating the significance tests on a larger writer sample for more statistical power, and testing under a writer-independent training protocol. The pipeline, locked experiment notebooks, and full results for both datasets are available in the `6buwan` repository.

References

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